

# Tag X3 BLE Sensor Tag & Beacon Datasheet

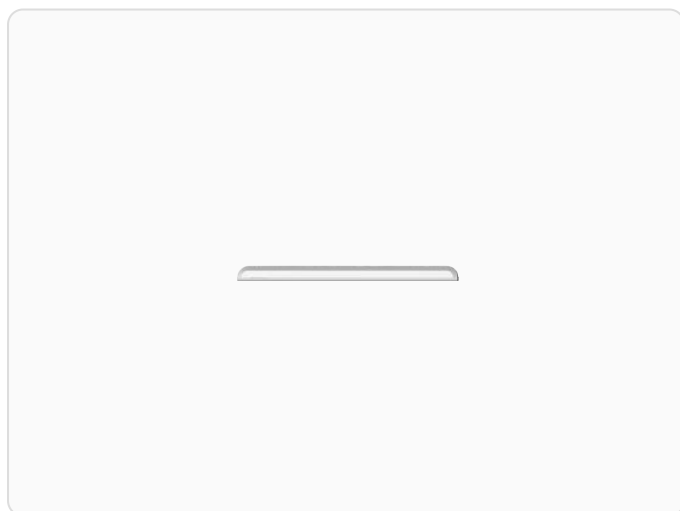
## Product Overview

**Tag X3** is a compact and rugged **Bluetooth Low Energy (BLE) sensor tag and beacon** designed for **asset tracking, indoor location services, and condition monitoring** across industrial and commercial environments. Supporting both **iBeacon** and **Eddystone** protocols, Tag X3 enables seamless identification, proximity detection, and location tracking using BLE gateways, mobile devices, and edge receivers.

In addition to beaconing, Tag X3 integrates multiple onboard sensors including an **IMU (motion and orientation sensing)**, **temperature**, **humidity**, and **barometric pressure**, allowing it to capture real-time movement and environmental conditions directly from the asset. This makes it ideal for monitoring **asset usage, movement patterns, handling events, and storage conditions**.

Engineered for **ultra-low power operation**, Tag X3 supports configurable advertising intervals, transmit power levels, and event-based triggers to maximize battery life while maintaining reliable visibility. The device can operate continuously for long durations in both **broadcast-only** and **sensor-enabled** modes.

Tag X3 works seamlessly with **BLE-to-Wi-Fi gateways, smartphones, tablets, and industrial edge devices**, enabling data forwarding to **on-premise systems or cloud platforms** for analytics, alerts, and automation. Its small form factor and rugged enclosure make it suitable for **tools, containers, equipment, pallets, and movable assets** in warehouses, factories, campuses, and smart buildings.





## Key Features

### BLE Beacon with iBeacon & Eddystone Support

Fully supports industry-standard iBeacon and Eddystone protocols for indoor positioning, proximity detection, and asset identification.

### Temperature, Humidity & Pressure Sensing

Onboard environmental sensors provide real-time monitoring of temperature, relative humidity, and barometric pressure.

### Configurable Advertising Parameters

Adjustable advertising interval, transmit power, and payload structure to balance range, visibility, and power consumption.

### Gateway & Smartphone Compatible

Works seamlessly with BLE gateways, smartphones, tablets, and industrial edge devices for data collection.

### BLE Configuration via Maya App

Easy local configuration, diagnostics, and parameter updates using the Macman Maya mobile application.

### Integrated IMU for Motion & Orientation

Built-in inertial sensor enables motion detection, tilt sensing, shock events, and movement-based triggers.

### Ultra-Low Power BLE Operation

Optimized BLE advertising and sensor sampling for long battery life in continuous broadcast and event-driven modes.

### Event-Based Sensor Triggers

Supports motion-based, orientation-based, and threshold-based triggers for intelligent tracking and alerts.

### Compact & Rugged Form Factor

Small, lightweight enclosure suitable for tools, pallets, containers, equipment, and movable assets.

### Indoor Asset Tracking & Condition Monitoring Ready

Designed for RTLS, asset tracking, usage analytics, and environmental condition visibility.

# Suited Applications

## Asset Tracking & RTLS

Real-time tracking of tools, equipment, pallets, containers, and movable assets using BLE gateways and indoor location systems.

## Smart Buildings & Campuses

Indoor location, proximity detection, and environmental monitoring across offices, campuses, and commercial buildings.

## Industrial Equipment Monitoring

Motion, shock, and usage detection for machines, tools, and equipment to improve asset utilization and maintenance planning.

## Retail & Indoor Analytics

Footfall analysis, zone activity tracking, and customer movement insights without using cameras.

## Warehouse & Logistics Operations

Asset visibility across warehouses, distribution centers, and logistics hubs to track movement, utilization, and dwell time.

## Cold Chain & Storage Condition Monitoring

Temperature, humidity, and pressure monitoring for sensitive goods during storage and indoor transportation.

## Personnel & Safety Monitoring

Proximity and movement tracking for workforce safety, zone compliance, and restricted area monitoring.

## Facility & Infrastructure Monitoring

Environmental condition visibility and asset movement tracking inside factories, plants, and infrastructure facilities.

# Technical Specifications

## General Specifications

|                   |                              |
|-------------------|------------------------------|
| Part Order Number | TagX3                        |
| Product Brand     | Macnman Technologies Pvt Ltd |

## Environmental Sensor Specifications

|                               |                                       |
|-------------------------------|---------------------------------------|
| Temperature Sensor            | Integrated digital temperature sensor |
| Temperature Measurement Range | -20°C to +60°C                        |
| Temperature Accuracy          | ±0.5°C (typical)                      |
| Humidity Sensor               | Integrated digital humidity sensor    |
| Humidity Measurement Range    | 0% to 100% RH (non-condensing)        |
| Humidity Accuracy             | ±3% RH (typical)                      |
| Pressure Sensor               | Integrated barometric pressure sensor |

|                            |                     |
|----------------------------|---------------------|
| Pressure Measurement Range | 300 hPa to 1100 hPa |
| Pressure Accuracy          | ±1 hPa (typical)    |

## IMU Specifications

|                    |                                                                                       |
|--------------------|---------------------------------------------------------------------------------------|
| IMU Type           | Integrated MEMS Inertial Measurement Unit                                             |
| Sensors Included   | 3-axis Accelerometer<br>3-axis Gyroscope (variant dependent)                          |
| Acceleration Range | ±2g / ±4g / ±8g / ±16g (configurable)                                                 |
| Motion Detection   | Movement detection<br>Shock & impact detection<br>Tilt & orientation change detection |
| Sampling Rate      | Configurable (Typical: 1 Hz to 100 Hz)                                                |
| Trigger Modes      | Event-based triggers<br>Threshold-based triggers<br>Periodic sampling                 |
| Use Cases          | Asset movement tracking<br>Drop / shock detection<br>Usage pattern analysis           |

## Wireless Transmission

|                            |                                                                                     |
|----------------------------|-------------------------------------------------------------------------------------|
| Wireless Technology        | Bluetooth Low Energy (BLE)                                                          |
| Supported Beacon Protocols | iBeacon<br>Eddystone (UID / URL / TLM)                                              |
| Operating Frequency        | 2.4 GHz ISM Band                                                                    |
| Transmit Power             | Configurable (Typical: -20 dBm to +4 dBm)                                           |
| Advertising Interval       | Configurable (100 ms to 10 s)                                                       |
| Antenna                    | Internal high-efficiency BLE antenna                                                |
| Communication Mode         | Broadcast (Beacon Advertising)<br>Connectable BLE (for configuration & diagnostics) |

## LED Status

|          |                            |
|----------|----------------------------|
| LED Type | 1 Internal Multi-Color LED |
|----------|----------------------------|

|                     |                                                |
|---------------------|------------------------------------------------|
| Solid Blue          | Device is connected with Maya App              |
| Flashing Blue       | OTA (Over-the-Air) update in progress          |
| Slow Blinking Green | Attempting to connect to the network           |
| Single Red Blink    | A confirmed parameter of tag is being modified |

## Device Configuration

|                      |                                               |
|----------------------|-----------------------------------------------|
| Configuration        | Via Macnman Maya Android App                  |
| Uplink Modes         | Time Based / Threshold Based / Sampling Based |
| Stored Records*      | 30,000 Records                                |
| Remote Configuration | JSON Format Data - from server                |

## Battery Performance

|                        |                                                                                                       |
|------------------------|-------------------------------------------------------------------------------------------------------|
| Battery Type           | CR Series Lithium Coin Cell (Replaceable)                                                             |
| Typical Battery Models | CR2477 / CR2450 (Variant dependent)                                                                   |
| Power Consumption      | Ultra-low power BLE operation<br>Optimized for beacon advertising and sensor event-based transmission |
| Battery Life           | Up to 12–24 months (depending on advertising interval, Tx power, and sensor usage)                    |
| Battery Replacement    | User-replaceable coin cell                                                                            |

## Supported BLE Protocols

|                   |                                                                                   |
|-------------------|-----------------------------------------------------------------------------------|
| Beacon Protocols  | iBeacon<br>Eddystone-UID<br>Eddystone-URL<br>Eddystone-TLM                        |
| BLE Communication | BLE Advertising (Broadcast Mode)<br>Connectable BLE (Configuration & Diagnostics) |
| Compatibility     | BLE Gateways<br>Smartphones & Tablets (Android / iOS)<br>Industrial Edge Devices  |

## Physical Characteristics

|                       |                                                                   |
|-----------------------|-------------------------------------------------------------------|
| Power Supply          | Internal CR Series Lithium Coin Cell                              |
| Operating Temperature | -20°C to +60°C                                                    |
| Operating Humidity    | 0% to 95% RH (non-condensing)                                     |
| Ingress Protection    | IP54 (IP65 optional variant)                                      |
| Housing Material      | Industrial-grade ABS / Polycarbonate enclosure                    |
| Color                 | Matte Black                                                       |
| Weight                | < 50 g (with battery)                                             |
| Mounting Options      | Adhesive mount / Screw mount / Lanyard / Clip (variant dependent) |

## Certifications & Compliance

|      |                            |
|------|----------------------------|
| WPC  | Compliant                  |
| RoHS | Conforms to RoHS directive |

## Security & Encryption

|                     |                                    |
|---------------------|------------------------------------|
| Data Encryption     | AES-256 end-to-end encryption      |
| Authentication      | OTAA activation modes              |
| Firmware Protection | Secure boot and signed OTA updates |

## Tag X3 Variants

| Variant Name | Power Source    | Battery Type                | Battery Capacity | External Flash | Ideal For                                                                                                              |
|--------------|-----------------|-----------------------------|------------------|----------------|------------------------------------------------------------------------------------------------------------------------|
| TagX3-BO     | Battery Powered | CR Series Lithium Coin Cell | CR2450 / CR2477  |                | Asset tracking, RTLS, condition monitoring, cold chain, warehouse logistics, smart buildings, and portable deployments |

## What's in the Box?

Make sure box contains following items before installations



**MacRay**



**Mounting  
Clip\***



**User manual &  
Warranty Card**



**IP  
Extender\***



**Mounting  
Screws**

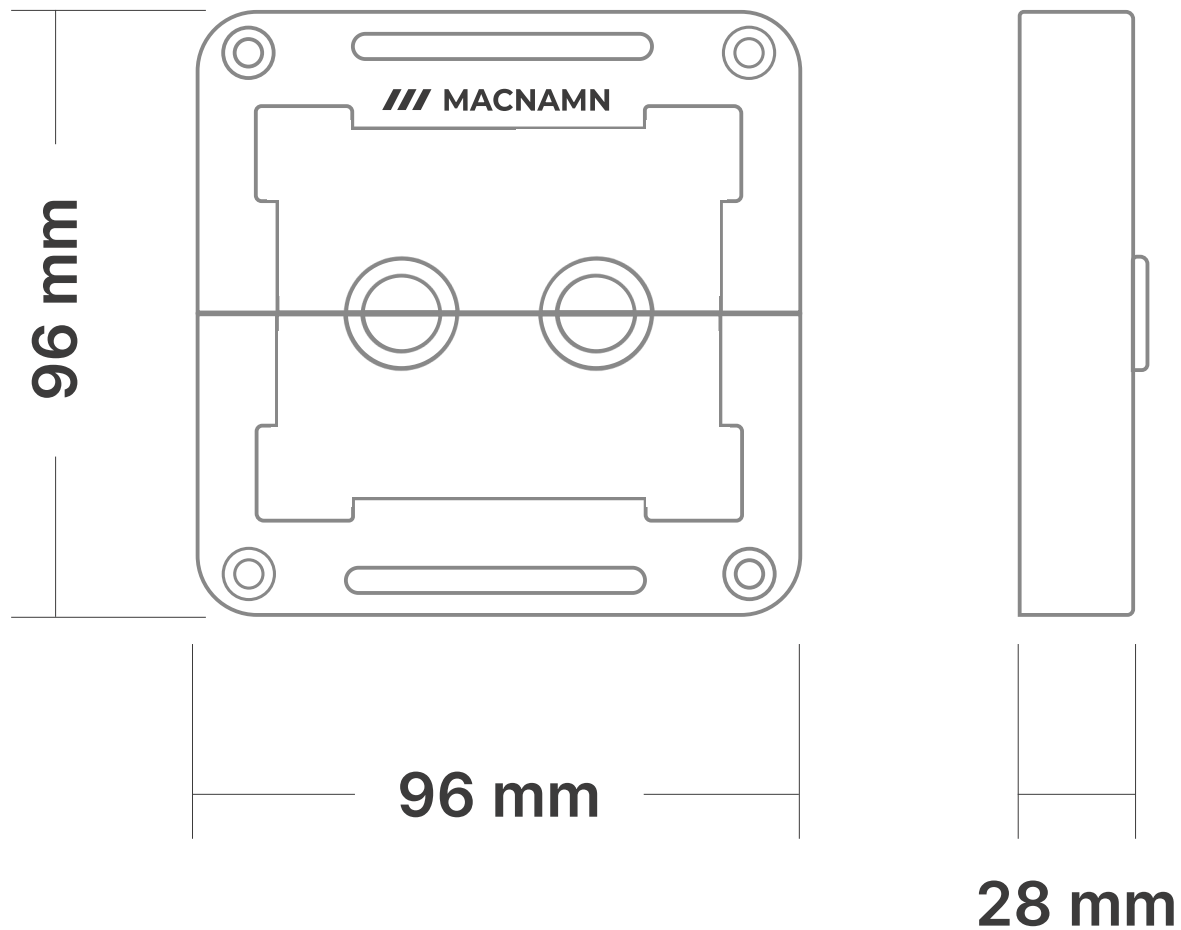


**Power  
Adapters\***

### **Note**

Accessories must be purchased separately & may not be included in the box.

## Tag X3 Dimensions



## Frequently Asked Questions

### What is the Tag X3 used for?

Tag X3 is a compact, rugged BLE sensor tag and beacon (iBeacon/Eddystone) for asset tracking, indoor location and condition monitoring with long battery life. Typical applications include Asset Tracking & RTLS, Warehouse & Logistics Operations, Smart Buildings & Campuses, Cold Chain & Storage Condition Monitoring, Industrial Equipment Monitoring and Personnel & Safety Monitoring.

## What does the Tag X3 measure, and how accurately?

Measurement specifications of the Tag X3:

- Temperature: range -20°C to +60°C, accuracy  $\pm 0.5^\circ\text{C}$  (typical).
- Humidity: range 0% to 100% RH (non-condensing), accuracy  $\pm 3\%$  RH (typical).
- Pressure: range 300 hPa to 1100 hPa, accuracy  $\pm 1$  hPa (typical).

## Which Bluetooth beacon protocols does the Tag X3 support?

The Tag X3 uses Bluetooth Low Energy (BLE) on the 2.4 GHz ISM Band. Supported beacon protocols: iBeacon, Eddystone (UID / URL / TLM). Transmit power: Configurable (Typical: -20 dBm to +4 dBm). Advertising interval: Configurable (100 ms to 10 s).

## How long does the Tag X3 battery last?

The Tag X3 battery type is CR Series Lithium Coin Cell (Replaceable). Typical battery models: CR2477 / CR2450 (Variant dependent). Battery life: Up to 12–24 months (depending on advertising interval, Tx power, and sensor usage).

## What is the IP rating and operating temperature of the Tag X3?

The Tag X3 has an ingress protection rating of IP54 (IP65 optional variant) and an operating temperature range of -20°C to +60°C. Operating humidity: 0% to 95% RH (non-condensing). Mounting options: Adhesive mount / Screw mount / Lanyard / Clip (variant dependent).

## Related Products and Resources

- Related datasheets: [Setu WX1 BLE to Wi-Fi gateway](#), [BLE Tag X2 industrial BLE asset tracker](#)
- Help: [Warranty terms and conditions](#) and [contact Macnman support](#)